

REAR SUSPENSION HELPER SPRING KIT

(DR1500DQ)

Premium Suspension Enhancement System • Installation Guide

SAFETY WARNINGS & CAUTIONS

- **Safety First:** Never work under a vehicle supported solely by a jack. Always secure the vehicle properly using heavy-duty jack stands on solid, level ground.
- **Personal Protection:** Always wear approved safety glasses and protective gloves to shield yourself from debris, corrosion, and sharp vehicle edges.
- **Load Limits:** Do not exceed the vehicle manufacturer's recommended gross vehicle weight rating (GVWR) or individual axle weight ratings.
- **Clearance Warning:** Under load, the rubber helper spring will expand in diameter. Ensure there is adequate clearance between the spring and any surrounding vehicle components (brake lines, wiring, chassis parts).

KIT COMPONENT DETAILS

ITEM	COMPONENT DESCRIPTION	QUANTITY (PER KIT)
1	Hex Bolt (Mounting Hardware)	2
2	Upper Mounting Locator / Spacer Disc	2
3	Premium Rubber Helper Spring	2
4	Lock Nut	2

Note: Depending on your model configuration, the helper spring assembly components may arrive pre-assembled.

STEP-BY-STEP INSTALLATION INSTRUCTIONS

- 1

Prepare the Vehicle

Park the vehicle on solid, level ground. Engage the parking brake and place wheel chocks around the tires that remain on the ground. Jack up the rear of the vehicle until the rear wheels are off the ground, then secure the chassis safely using heavy-duty jack stands.
- 2

Remove Factory Bump Stops

Locate the factory bump stop mounted in the steel cup on the vehicle frame. Using a flathead screwdriver or a small pry bar, carefully pry the factory rubber bump stop out of its mounting cup and discard it.
- 3

Clean and Lubricate

Clean the inside of the mounting cup thoroughly using warm, soapy water and a clean rag to remove all dirt, road debris, and corrosion. Once cleaned, apply a generous amount of soapy water inside the cup and around the upper rubber locator/spacer of your new helper spring assembly to act as a lubricant.

4 Install the Helper Spring

Align the upper rubber locator of the helper spring assembly with the open frame cup. Push and twist the helper spring assembly upward into the cup until the upper locator is fully engaged and seated inside.

5 Seat the Spring Assembly

Place a temporary spacer (such as a short, solid wooden block) on the axle pad directly beneath the helper spring. Carefully lower the vehicle jack until the weight of the vehicle rests on the spacer, which will press the helper spring completely and securely into the frame cup. Raise the vehicle back up to remove the spacer block.

6 Verify Clearance & Lower

Lower the vehicle completely to the ground so that the suspension rests in its normal, unloaded state. Verify that there is a clearance gap of 1/2" to 1-1/2" (13 to 38 mm) between the bottom of the helper spring and the contact point on the axle. This gap is necessary to maintain original ride quality when the vehicle is unloaded.

CLEARANCE AND OPERATION GUIDELINES

Unloaded State: For light trucks, SUVs, and vans, a physical gap of 1/2" to 1-1/2" is standard when the vehicle is empty. This prevents the helper spring from engaging over minor road bumps when empty, ensuring a smooth factory ride. A partial load or heavy aftermarket vehicle accessories will naturally reduce or eliminate this gap, which is expected and allows the spring to begin assisting the suspension under load.